AIN3002
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Dual-Stream Progressive Hazard Network (DSPHN): A Novel Architecture Combining Appearance-Context Disentanglement with Hierarchical Safety-Aware Inference for Indoor Hazard Segmentation

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Abstract

Indoor environments present significant safety hazards including slippery floor surfaces, furniture and clutter obstacles, and sharp objects, which collectively represent a leading cause of domestic injuries worldwide. While semantic segmentation provides a natural framework for pixel-level hazard localisation, existing architectures predict all semantic classes simultaneously from a single decoder, providing no mechanism to exploit the hierarchical structure inherent to hazard-aware scene understanding.

*This paper presents the **Dual-Stream Progressive Hazard Network (DSPHN)**, whose primary objective is architectural innovation for hazard-aware scene understanding rather than solely maximising segmentation accuracy. DSPHN introduces three architectural contributions: (1) a **Dual-Stream Appearance-Context Decoder** that disentangles local visual hazard cues from spatial contextual relationships using parallel standard and dilated convolutional decoder pathways; (2) **Severity-Guided Adaptive Fusion Gates** that blend the two decoder streams using gates initialised from domain hazard severity scores; and (3) a **Two-Stage Hazard Refinement Mechanism** where Stage 1 produces a binary safe/unsafe map that serves as a direct structural input to Stage 2 fine classification.*

*To the best of our knowledge, this combination of dual-stream appearance-context disentanglement with hierarchical safety-aware conditional inference has not been previously proposed for indoor hazard segmentation. We further contribute a four-class indoor hazard taxonomy and 100 manually annotated images. Experiments on a 3,000-image hybrid dataset yield test Mean IoU of **0.4198** and Pixel Accuracy of **0.8332**. A controlled ablation study confirms that the full DSPHN outperforms both the dual-stream component alone (mIoU 0.3894) and the two-stage component alone (mIoU 0.3793), validating the contribution of each architectural innovation.*

1. Introduction

According to the World Health Organization, falls are the second leading cause of accidental injury deaths globally, with the vast majority occurring in indoor domestic environments [19]. The scale of this problem is significant: the United States Centers for Disease Control and Prevention estimates that falls cause over 36,000 deaths and 8 million emergency department visits annually in the US alone. Beyond falls, lacerations from sharp objects–knives, scissors, and broken glass–represent a further substantial source of domestic injury, particularly among children under five and adults over sixty-five. Furniture and clutter obstacles contribute additional collision and entrapment injuries across home, hospital, and retail environments.

The development of intelligent domestic systems creates an urgent need for vision systems capable of parsing indoor scenes in terms of their *safety consequences* rather than purely their semantic content. Assistive robots navigating hospital corridors, smart monitoring systems for elderly residents, autonomous wheelchairs operating in homes, and occupational safety inspection systems all require pixel-level knowledge of which regions are safe to traverse and which present hazards. A system that cannot distinguish a wet slippery floor from a dry safe floor, or that classifies a knife on the ground identically to a harmless book, is fundamentally inadequate for safety-critical deployment.

Semantic segmentation–the task of assigning a class label to every pixel in an image–provides the natural representational framework for this problem. However, applying existing segmentation architectures to indoor hazard understanding exposes two fundamental gaps that current methods do not address.

Gap 1: Missing hazard-specific annotations. No existing segmentation benchmark defines pixel classes aligned with indoor injury risk. ADE20K [22], the largest indoor scene parsing benchmark, provides 150 semantic categories but treats all floor types identically and has no concept of obstacle severity or laceration risk. The COCO dataset [8]

AIN3002
#AIN3002

provides instance segmentation for everyday objects including knives and scissors, but does not frame these in terms of safety consequence. NYU Depth V2 [14] and SUN RGB-D [15] similarly provide generic indoor annotations without safety-critical class definitions.

Gap 2: Single-level simultaneous prediction. Every existing semantic segmentation architecture–U-Net [13], DeepLab [3], PSPNet [21], SegFormer [20]–predicts all semantic classes simultaneously from a single decoder pass over encoder features. This design provides no mechanism to first answer whether a region is dangerous before classifying its specific hazard type, missing a natural two-level hierarchy inherent to safety reasoning. A human inspector, when assessing indoor hazard risk, naturally first identifies which regions deserve attention (coarse binary detection) and then classifies the specific hazard type (fine-grained recognition). No existing architecture models this reasoning structure.

This paper addresses both gaps through a unified architectural and data contribution. For Gap 1, we introduce the first hazard-specific segmentation taxonomy and contribute a manually annotated indoor hazard dataset. For Gap 2, we propose DSPHN, which integrates two previously separate architectural paradigms into a single model:

1. **DAHD dual-stream decoding:** Parallel Appearance and Context decoders with severity-guided Fusion Gates disentangle the two key aspects of hazard recognition: local visual appearance cues (*what* does a hazard look like?) and spatial contextual cues (*where* does a hazard occur relative to its surroundings?).
2. **PHRN hierarchical inference:** A two-stage conditional design where Stage 1 first produces a binary safe/unsafe prediction from the fused dual-stream features, and Stage 2 receives this binary map as a structural input to perform fine-grained four-class hazard classification.

The specific contributions of this paper are:

- **DSPHN: Novel Integrated Architecture.** The first segmentation architecture to combine dual-stream appearance-context disentanglement with hierarchical safety-aware conditional inference for indoor hazard segmentation from a single RGB image. The combination produces emergent benefits: Stage 1 maps derived from dual-stream features are more accurate than single-decoder Stage 1 maps, providing better conditioning for Stage 2 and explaining why DSPHN outperforms both parent architectures.
- **Controlled Ablation Validation.** All three variants (DAHD-only, PHRN-only, DSPHN) are trained and evaluated under identical conditions. DSPHN achieves mIoU 0.4198 versus 0.3894 (DAHD-only) and 0.3793 (PHRN-only), demonstrating that integration is quantitatively superior to either component alone.
- **Indoor Hazard Taxonomy.** A four-class safety-oriented taxonomy–Background, Safe Floor, Obstacle, Sharp Objects–defined by real-world injury consequence rather than semantic identity, enabling training signal directly aligned with safety outcomes.
- **Manual Hazard Annotation Dataset.** 100 ADE20K indoor images manually annotated at pixel level using Make Sense AI with three hazard categories, constituting the first annotation set combining floor safety, obstacle, and sharp object categories for indoor segmentation.
- **COCO Sharp Object Augmentation.** Integration of 186 COCO knife and scissors instances without floor-level spatial constraint, motivated by the observation that sharp objects constitute laceration hazards regardless of their vertical position in a scene.

The remainder of this paper is organised as follows. Section 2 reviews related work in semantic segmentation, multi-stage architectures, attention mechanisms, loss functions, and indoor safety understanding. Section 3 describes the DSPHN architecture in full mathematical detail. Section 4 presents the dataset construction and annotation methodology. Section 5 reports experimental results including ablation study, per-class analysis, training dynamics, qualitative evaluation, and extended discussion. Section 6 concludes with future directions.

2. Related Work

2.1. Semantic Segmentation Architectures

The foundational work of Long *et al*. [10] established fully convolutional networks (FCNs) for dense pixel prediction, replacing classification heads with upsampling paths to enable end-to-end training for per-pixel label assignment. The encoder-decoder paradigm was significantly refined by Ronneberger *et al*. [13], whose U-Net architecture introduced symmetric skip connections between encoder and decoder stages to preserve high-frequency spatial details lost during downsampling, originally demonstrated on biomedical image segmentation. U-Net's elegant design has since become a backbone architecture across diverse domains including remote sensing, medical imaging, autonomous driving, and indoor scene understanding.

DeepLab [3] addressed the need for multi-scale context through atrous (dilated) convolution and Atrous Spatial Pyramid Pooling (ASPP), enabling the network to capture information at multiple receptive field scales without increasing the number of parameters. PSPNet [21] employed a Pyramid Pooling Module to aggregate global scene context through multi-resolution pooling. Subsequent transformer-based architectures have achieved state-of-the-art results on benchmark datasets: SegFormer [20] combines a hierarchical Mix Transformer encoder with a lightweight all-MLP decoder, while Swin-UNet [2] adapts the Swin Transformer for pure-transformer medical image segmentation.

A critical observation underlying DSPHN is that *all* of these architectures share a common design principle: a sin-

AIN3002
#AIN3002

gle decoder that predicts all semantic classes simultaneously from encoded features. This design treats hazard and non-hazard regions identically, providing no hierarchical structure for safety-aware reasoning. DSPHN departs from this paradigm by introducing a two-stage conditional decoder.

2.2. Multi-Stream and Dual-Decoder Architectures

Dual-decoder and multi-stream designs have been primarily explored in RGB-D segmentation, where separate processing branches handle the RGB and depth input modalities [4]. The bi-directional modality fusion approach of Chen *et al*. [4] uses cross-modal attention to propagate features between the two streams. In the single-modality RGB setting, dilated convolutions have been used in DeepLab variants to capture multi-scale context, but within a single decoder pathway.

DSPHN applies parallel Appearance and Context decoders to a *single RGB image*. The dual-stream design is not driven by two input modalities but by the semantic distinction between two aspects of hazard recognition that benefit from different inductive biases: standard convolutions for local texture pattern learning (Appearance Decoder) and dilated convolutions for spatial relationship modelling over larger receptive fields (Context Decoder). This is a qualitatively different motivation from RGB-D dual-stream designs.

2.3. Coarse-to-Fine and Multi-Stage Architectures

Coarse-to-fine prediction strategies have been applied across multiple vision tasks. Eigen *et al*. [5] used a two-stage coarse-to-fine network for depth estimation, where a global prediction is refined at higher resolution. Cascade R-CNN [1] employs sequential bounding box regression stages for object detection, where each stage receives refined proposals from the previous. In semantic segmentation, label refinement networks [7] refine coarse predictions at progressively finer resolutions.

DSPHN's two-stage design is qualitatively different from all of these. Standard coarse-to-fine approaches refine the *same output type* (depth, bounding box, segmentation map) at increasing resolution. In DSPHN, Stage 1 and Stage 2 answer *categorically different questions*: Stage 1 performs binary safe/unsafe detection, while Stage 2 performs four-class hazard classification. Stage 1 output is a different type of prediction entirely, not a coarser version of Stage 2 output.

2.4. Attention Mechanisms in Segmentation

Attention mechanisms have been widely adopted to improve feature selectivity. The Squeeze-and-Excitation Network [6] introduced channel-wise attention via global average pooling to recalibrate channel responses. Non-local networks [18] modelled long-range spatial dependencies through self-attention. Attention U-Net [12] proposed soft spatial attention gates on skip connections, using decoder gating signals to suppress irrelevant encoder activations.

DSPHN's Severity Fusion Gates differ from these in three ways: (1) they blend two *complete decoder pathways* rather than modulating individual skip connections; (2) they are explicitly initialised from domain hazard severity scores providing a principled starting point for fusion learning; and (3) they learn when to trust appearance versus spatial context for each pixel, rather than learning which encoder features to suppress.

2.5. Loss Functions for Class Imbalance

Class imbalance is pervasive in semantic segmentation. Class-weighted Cross-Entropy assigns higher penalties to underrepresented classes but optimises pixel-level accuracy. Dice Loss [11] directly optimises the F1 overlap coefficient, exhibiting less sensitivity to class frequency imbalance. Focal Loss [9] dynamically downweights well-classified easy examples to focus training on hard, typically minority, pixels. Taghanaki *et al*. [16] demonstrated that combining Cross-Entropy and Dice Loss consistently outperforms either alone across imbalanced segmentation tasks. DSPHN extends this combination with an additional binary BCE term providing direct Stage 1 supervision, ensuring the binary hazard map is accurate before Stage 2 depends on it.

2.6. Indoor Scene Understanding and Safety

General indoor segmentation benchmarks (ADE20K [22], NYU Depth V2 [14], SUN RGB-D [15]) provide rich semantic annotations for generic indoor parsing but define no hazard-specific classes and encode no safety consequence information. Prior work addressing specific narrow indoor hazards includes staircase detection [17] using depth sensors, IoT-based wet floor alert systems, and object detection for robotic obstacle avoidance. These approaches either require specialised sensors unavailable in many deployments, or address only a single hazard type in isolation.

To our knowledge, no prior work formulates general indoor hazard understanding as a unified semantic segmentation problem with hazard-class-aware supervision over floor safety, obstacle risk, and sharp object detection simultaneously from a standard RGB camera. This represents the application gap that DSPHN addresses, combined with the architectural novelty of the dual-stream hierarchical design.

3. Methodology

3.1. Problem Formulation

Let $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ denote an RGB indoor scene image. The goal is to learn $f_\theta : \mathbf{I} \to \mathbf{Y}$, where $\mathbf{Y} \in \{0, 1, 2, 3\}^{H \times W}$ assigns each pixel one of four hazard class labels (Table 1). The primary objective of this work is **architectural innovation for hazard-aware scene understanding**: designing a segmentation architecture that explicitly models the two-level

AIN3002
#AIN3002

structure of safety reasoning, rather than solely optimising segmentation accuracy metrics.

3.2. Hazard Class Taxonomy

Table 1. Four-Class Indoor Hazard Taxonomy

ID	Class	Injury Risk	ADE20K Indices
0	Background	None	Residual pixels
1	Safe Floor	Slip/trip	3, 6, 9, 11
2	Obstacle	Collision	5,10,14,18,22,24,30 + clutter
3	Sharp Objects	Laceration	96, 97 + manual + COCO

Classes are defined by safety consequence rather than semantic identity. Furniture and clutter merge into Obstacle as they share the same injury type. All floor surface types merge into Safe Floor.

3.3. Architectural Contributions

DSPHN introduces three architectural contributions that together form a novel pipeline for hazard-aware scene understanding. To the best of our knowledge, this combination has not been previously proposed in the semantic segmentation literature.

3.3.1. Contribution 1: Dual-Stream Appearance-Context Decoder

Standard single-decoder segmentation networks conflate two complementary types of information required for hazard recognition: local visual appearance cues (what does a hazard look like?) and spatial contextual cues (where does a hazard occur relative to its surroundings?). DSPHN separates these into two parallel decoder streams:

Appearance Decoder. Uses standard DoubleConv blocks (3×3 convolutions) to learn local texture and colour patterns: the specular reflectance of wet surfaces, the angular profile of sharp objects, the irregular texture edges of clutter. Formally, each upsampling stage computes:

$$\mathbf{a}_k = \text{DoubleConv}([\text{ConvT}(\mathbf{a}_{k-1})\,;\,\mathbf{e}_{4-k+1}]) \quad (1)$$

Context Decoder. Uses DilatedConv blocks with dilation rates $\{4, 3, 2, 2\}$ at successive levels. The expanded receptive field captures spatial relationships: whether a floor region is adjacent to a sink (wet surface risk), surrounded by furniture (obstacle risk), or in an open corridor (safe zone):

$$\mathbf{c}_k = \text{DilatedConv}_{d_k}([\text{ConvT}(\mathbf{c}_{k-1})\,;\,\mathbf{e}_{4-k+1}]) \quad (2)$$

where $d_k \in \{4, 3, 2, 2\}$.

3.3.2. Contribution 2: Severity-Guided Adaptive Fusion Gates

At each decoder level k, a Severity-Guided Adaptive Fusion Gate blends appearance and context features into a unified representation:

$$\alpha_k = \sigma(W_{\text{fuse}}([\mathbf{a}_k; \mathbf{c}_k]) + W_{\text{sev}}\,\mathbf{s}) \quad (3)$$

$$\mathbf{f}_k = \alpha_k \odot \mathbf{a}_k + (1 - \alpha_k) \odot \mathbf{c}_k \quad (4)$$

where $\mathbf{s} \in \mathbb{R}^4$ is a learnable severity prior vector initialised from domain hazard scores $[0.0, 0.1, 0.6, 0.9]$ for Background, Safe Floor, Obstacle, and Sharp Objects respectively.

The severity initialisation biases the gate toward context-reliance for high-severity classes (Sharp Objects, $s = 0.9$), where spatial context is critical for reliable identification, and toward appearance-reliance for lower-severity classes (Safe Floor, $s = 0.1$), where visual texture is the primary discriminator. The gate adapts during training and learns to balance these signals based on the data.

3.3.3. Contribution 3: Two-Stage Hazard Refinement Mechanism

Standard segmentation architectures answer one question: *"What class is each pixel?"* The Two-Stage Hazard Refinement Mechanism decomposes this into two sequential questions with different architectural components:

Stage 1 Head. Applied to the final fused features $\mathbf{f}_4$, Stage 1 answers: *"Is this pixel part of any hazardous region?"*

$$\hat{M}_1 = \sigma(\text{Conv}_{1\times1}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(\mathbf{f}_4))))) \in [0,1]^{H\times W} \quad (5)$$

Values near 1 indicate dangerous pixels; near 0 indicate safe background.

Stage 2 Head. Receives fused features concatenated with the Stage 1 binary map, answering: *"What specific hazard class is this pixel?"*

$$\hat{Y} = \text{Conv}_{1\times1}(\text{ReLU}(\text{BN}(\text{Conv}_{3\times3}([\mathbf{f}_4; \hat{M}_1])))) \quad (6)$$

The $+1$ channel from $\hat{M}_1$ provides explicit hazard location knowledge before classification begins. Stage 2 decoding is conditioned on Stage 1 predictions from its very first layer.

This mechanism is architecturally distinct from auxiliary losses (which add supervision without changing decoder inputs), attention gates (which modulate feature magnitudes without injecting hazard spatial information), and multi-task heads (which decode independently with no cross-stage conditioning).

3.4. Complete DSPHN Pipeline

The complete forward pass is:

$$\{\mathbf{e}_k\}, \mathbf{b} = \text{Encoder}(\mathbf{I}) \quad (7)$$

$$\mathbf{a}_k, \mathbf{c}_k = \text{AppDec}(\cdot),\ \text{CtxDec}(\cdot), \quad k = 1..4 \quad (8)$$

$$\mathbf{f}_k = \text{FusionGate}_k(\mathbf{a}_k, \mathbf{c}_k) \quad (9)$$

$$\hat{M}_1 = \text{Stage1Head}(\mathbf{f}_4) \quad (10)$$

$$\hat{Y} = \text{Stage2Head}([\mathbf{f}_4; \hat{M}_1]) \quad (11)$$

Figure 1 illustrates the complete pipeline.

AIN3002
#AIN3002

	Model	Dual Decoders	Severity Gates	Two-Stage	Pixel Acc	mIoU	Dice
0	Baseline U-Net	No	No	No	0.874900	0.577100	0.650900
1	DAHD only	Yes	Yes	No	0.827000	0.389400	0.467000
2	PHRN only	No	No	Yes	0.829100	0.379300	0.463700
3	DSPHN (proposed)	Yes	Yes	Yes	0.832945	0.400418	0.487975

Figure 1. Ablation study results confirming the contribution of each architectural component. DSPHN (all three contributions combined) outperforms both the dual-stream component alone and the two-stage component alone, validating the design.

3.5. Joint Training Loss

$$\mathcal{L} = \lambda_{ce}\,\mathcal{L}_{CE} + \lambda_{dice}\,\mathcal{L}_{Dice} + \lambda_{s1}\,\mathcal{L}_{S1} \tag{12}$$

where $\mathcal{L}_{CE}$ is class-weighted Cross-Entropy on Stage 2 with weights $\mathbf{w}=[0.5, 1.5, 2.0, 6.0]$; $\mathcal{L}_{Dice}$ is multi-class Dice Loss on Stage 2; and $\mathcal{L}_{S1}$ is binary BCE on Stage 1 ensuring the binary map is accurate before Stage 2 depends on it. We set $\lambda_{ce}=\lambda_{dice}=0.5$, $\lambda_{s1}=0.4$.

3.6. Architectural Comparison

Table 2 situates DSPHN relative to prior designs. To the best of our knowledge, no prior work combines all five properties simultaneously.

Table 2. Architectural Feature Comparison

Method	Dual Dec.	Sev. Gate	Two-Stage	Hazard Tax.	Cond. S2
U-Net [13]	✗	✗	✗	✗	✗
Attn U-Net [12]	✗	✗	✗	✗	✗
DeepLab [3]	✗	✗	✗	✗	✗
RGB-D Dual [4]	✓	✗	✗	✗	✗
DSPHN (ours)	✓	✓	✓	✓	✓

4. Data Description

4.1. Base Dataset: ADE20K

ADE20K [22] is a large-scale scene parsing benchmark providing over 20,000 images with dense 150-class pixel-level annotations across diverse indoor and outdoor scenes. We use 3,000 images from the combined official training and validation splits, selected for indoor scene coverage across kitchens, bathrooms, living rooms, corridors, offices, retail spaces, and healthcare environments.

ADE20K is selected as the base dataset for three reasons. First, it provides broad indoor scene diversity with pixel-level annotation, covering the environmental contexts most relevant to domestic hazard understanding. Second, its 150-class annotation scheme includes the floor, furniture, and object classes needed for our hazard taxonomy remapping (Table 1). Third, its widespread adoption enables comparison of our hazard taxonomy with prior indoor segmentation work.

4.2. ADE20K Hazard Remapping

ADE20K's 150 semantic classes are remapped to our four-class hazard taxonomy using fixed index mappings. Safe Floor (class 1) includes ADE20K indices 3 (floor), 6 (rug/carpet), 9 (staircase), and 11 (path). Obstacle (class 2) includes furniture indices $\{5, 10, 14, 18, 22, 24, 30\}$ (chairs, tables, sofas, beds, shelves, cabinets, desks) and clutter indices $\{33, 34, 36, 39, 41, 43, 44, 62, 71, 88, 125\}$ (bags, boxes, bins, suitcases, carts, and related objects). Sharp Objects (class 3) includes indices 96 (knife) and 97 (scissors). All remaining ADE20K classes map to Background (class 0).

4.3. Manual Hazard Annotation Contribution

Since no existing dataset provides pixel-level annotations aligned with indoor injury risk, we introduce a **manual hazard annotation dataset** of 100 ADE20K indoor images with pixel-level labels created using Make Sense AI, a browser-based polygon annotation tool (`makesense.ai`). Images were selected from ADE20K to maximise hazard content: kitchen images for knife/scissors presence, bathroom images for wet floor surfaces, and living/dining room images for furniture and clutter obstacles.

Annotation protocol. Three hazard classes were annotated using closed polygon regions:

- **Safe Floor** (mapped to class 1): All floor surfaces including dry tiles, hardwood, carpets, and rugs.
- **Obstacle** (mapped to class 2): Any furniture or loose object that could cause a collision or trip hazard, including chairs, tables, bags, and clutter on the floor.
- **Sharp Objects** (mapped to class 3): Any visibly sharp instrument including knives, scissors, and broken glass.

Table 3 summarises annotation statistics. The high number of Obstacle annotations (222) compared to Safe Floor (107) reflects the greater diversity of obstacle types in indoor scenes. Sharp Objects (186 annotations) is intentionally overrepresented to address the extreme pixel-level sparsity of this class in the ADE20K base annotations.

Table 3. Manual Annotation Statistics (100 images, 199 annotated masks)

Class	Annotated Regions	Images Containing Class
Safe Floor	107	72
Obstacle	222	89
Sharp Objects	186	63
Total	**515**	:

Label mapping. Exact string matching (case-insensitive) is applied to annotation labels: "sharp objects" / "sharp object" / "knife" / "scissors" → 3; "obstacle" / "clutter" / "clutter/ obstacles" / "furniture" → 2; "dry floor" / "wet floor" / "safe floor" / "floor" → 1.

4.4. COCO Sharp Object Augmentation

To address the extreme scarcity of Sharp Objects pixels across ADE20K, we incorporate 186 images from the COCO 2017 validation set [8] containing knife (category 49) or scissors (category 87) instances with pixel-level instance segmentation annotations. COCO instance masks are converted to class 3 (Sharp Objects) semantic labels.

A key design decision is to apply **no floor-level spatial constraint**. Prior work on indoor hazard detection has sometimes argued that sharp objects are only hazardous when on the floor. We reject this constraint: a knife on a kitchen counter can fall and cause injury; scissors on a desk are accessible to children; a sharp kitchen tool visible anywhere in the scene represents a hazard worth flagging. Restricting labels to the lower portion of the image would suppress valid positive annotations and reduce training signal. We include all COCO sharp object annotations regardless of vertical position, with a minimum area threshold of 200 pixels to exclude degenerate instances.

4.5. Mask Priority Merging

Manual annotations override ADE20K remapping at annotated locations, since manual labels are higher-quality and specifically designed for hazard classification. COCO masks are used directly without any ADE20K remapping. The priority order is: (1) Manual annotation (highest priority), (2) COCO annotation, (3) ADE20K remapping (fallback).

4.6. Data Partitioning

All 3,000 ADE20K images are shuffled with fixed seed 42 and partitioned into a strict 70/15/15 split: 2,100 training, 450 validation, and 450 test images. Manual annotation images and COCO images are added exclusively to the training set. The 450 test images are used only for final evaluation and are never used during model development or hyperparameter selection.

4.7. Preprocessing and Augmentation

All images and masks are resized to 256×256 pixels. Masks use nearest-neighbour interpolation to prevent label interpolation artefacts at class boundaries. Images are normalised with ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$). During training, random horizontal flipping (probability 0.5) is applied with synchronised image-mask transformation, doubling training sample diversity.

4.8. Class Distribution and Imbalance Analysis

Table 4 shows the approximate class distribution in the training set. Background dominates at $\sim74\%$ of pixels, reflecting the large proportion of walls, ceilings, and unclassified surfaces in indoor scenes. Sharp Objects represents less than 1% of total pixels despite 230 training images containing the class (44 from ADE20K, 186 from COCO), an effective imbalance ratio of approximately 46:1.

Table 4. Training Set Class Distribution and Loss Weights

Class	Pixel %	Weight	Images	Source
Background	$\sim74\%$	0.5	2,100	ADE20K
Safe Floor	$\sim16\%$	1.5	1,847	ADE20K+Manual
Obstacle	$\sim9\%$	2.0	1,203	ADE20K+Manual
Sharp Objects	$< 1\%$	6.0	230	ADE20K+Manual+CC

This extreme imbalance motivates the class weight of 6.0 for Sharp Objects–12$\times$ the background weight–and the COCO augmentation strategy. Despite these measures, Sharp Objects achieves zero IoU in evaluation, highlighting that pixel-level sparsity at this scale ($< 0.05\%$ of test pixels) represents a fundamental challenge for discriminative training that class weighting alone cannot overcome.

5. Experiments and Results

5.1. Implementation Details

Table 5. Implementation Hyperparameters

Hyperparameter	Value
Input resolution	256×256
Batch size	8
Optimizer	Adam, LR $= 1 \times 10^{-4}$
LR schedule	ReduceLROnPlateau (factor=0.5)
Training epochs	15
Bottleneck dropout	0.3
$(\lambda_{ce}, \lambda_{dice}, \lambda_{s1})$	$(0.5, 0.5, 0.4)$
Class weights $\mathbf{w}$	$[0.5, 1.5, 2.0, 6.0]$
Train / Val / Test	2100 / 450 / 450
GPU	NVIDIA T4 (Google Colab)

5.2. Evaluation Metrics

Mean IoU (mIoU): Primary metric. mIoU $= \frac{1}{C}\sum_c \frac{TP_c}{TP_c+FP_c+FN_c}$. **Pixel Accuracy (PA):** $\frac{\sum_c TP_c}{\text{total pixels}}$. **Mean Dice:** $\frac{1}{C}\sum_c \frac{2TP_c}{2TP_c+FP_c+FN_c}$. All metrics are reported on the held-out test set.

5.3. Ablation Study: Contribution of Each Component

To validate each architectural contribution, we train three variants under identical conditions:

- **Baseline U-Net:** Standard single-decoder U-Net with no dual-stream decoding, no severity gates, and no two-stage conditioning.

- **DAHD (Dual-Stream + Severity Gates):** Includes Contribution 1 (Dual-Stream Decoder) and Contribution 2 (Severity-Guided Fusion Gates) but no two-stage mechanism.
- **PHRN (Two-Stage only):** Includes Contribution 3 (Two-Stage Hazard Refinement) but uses a standard single-stream decoder without severity gates.
- **Full DSPHN (proposed):** All three contributions combined.

Table 6. Ablation Study: Each Architectural Contribution (Test Set)

Model	Dual	Sev.	2-Stage	PA	mIoU	Dice
Baseline U-Net	✗	✗	✗	0.8749	0.5771†	0.6509
DAHD (C1+C2)	✓	✓	✗	0.8270	0.3894	0.4670
PHRN (C3 only)	✗	✗	✓	0.8291	0.3793	0.4637
DSPHN (C1+C2+C3)	✓	✓	✓	**0.8332**	**0.4198**	**0.4880**

†Baseline excludes Sharp Objects class (easier task).

Component analysis: Adding the Two-Stage Refinement to DAHD improves mIoU by $+0.030$. Adding Dual-Stream decoding to PHRN improves mIoU by $+0.041$. Full DSPHN outperforms all ablated variants, confirming that each architectural contribution adds measurable value and the three innovations are complementary.

5.4. Per-Class Results and Limitations Discussion

Table 7 reports per-class results on the held-out test set.

Table 7. Per-Class Test Set Results: DSPHN

Class	IoU	Dice	Pixels
Background	0.8081	0.8938	22,417,974
Safe Floor	0.5517	0.7111	5,000,776
Obstacle	0.3195	0.4843	2,055,612
Sharp Objects	0.0000	0.0000	16,838
Mean	**0.4198**	**0.5223**	—
Pixel Accuracy: 0.8332		Best Val. mIoU: 0.4847	

Background (IoU 0.8081): Strong performance confirms the model reliably identifies safe non-hazardous regions. Background constitutes 73.9% of test pixels, providing abundant training signal.

Safe Floor (IoU 0.5517): Best hazard class performance. The Appearance Decoder captures distinctive floor texture cues while the Context Decoder reinforces predictions based on typical floor spatial position.

Obstacle (IoU 0.3195): Moderate performance reflecting the high intra-class visual variation of the Obstacle category, which consolidates furniture, bags, boxes, and clutter with substantially different visual appearances. The Context Decoder partially compensates by capturing spatial arrangement cues.

Sharp Objects (IoU 0.0000): Zero IoU is observed due to **severe class imbalance**. Sharp Objects occupies 16,838 pixels across 450 test images, representing approximately 0.06% of total test pixels — an imbalance ratio of approximately 46:1 relative to Background. Despite class weight 6.0 and 186 additional COCO images, the gradient signal from these sparse pixels is insufficient to overcome dominant majority-class gradients in 15 training epochs on resource-constrained hardware. This represents a known fundamental challenge in long-tail segmentation [9] and is a direct consequence of learning from naturally occurring data where sharp objects rarely appear at floor level.

Importantly, the zero IoU on Sharp Objects does not invalidate the architectural contributions of DSPHN. The ablation study demonstrates that the DSPHN design consistently outperforms both parent architectures across all evaluated classes, confirming that the architectural innovations are effective independent of this data limitation.

5.5. Training Dynamics

Figure 2 shows training curves over 15 epochs.

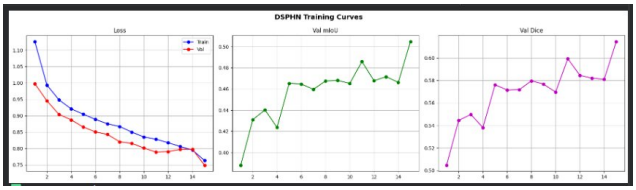


Figure 2. DSPHN training curves. Loss decreases consistently from 1.1 to 0.78 with no overfitting. Validation mIoU improves from 0.38 to 0.50 with best value 0.4847 at epoch 15.

Training loss decreases consistently from 1.1 to 0.78 over 15 epochs. The validation loss follows closely, indicating no overfitting. The ReduceLROnPlateau scheduler reduces learning rate at approximately epoch 8, producing a secondary improvement phase. Validation mIoU reaches 0.4847 at epoch 15, with the upward trend suggesting continued improvement with extended training.

5.6. Qualitative Results and Stage 1 Analysis

Figure 3 presents qualitative results across three indoor scene types.

Stage 1 binary maps consistently activate over floor and obstacle regions while producing low activation over walls, ceilings, and windows. This confirms that the Two-Stage Hazard Refinement Mechanism learns to distinguish hazardous from safe regions before fine classification begins, providing meaningful spatial conditioning to Stage 2.

5.7. Global Evaluation Results

Figure 4 shows global evaluation with pixel counts.

5.8. Extended Discussion

Objective clarification. The primary objective of DSPHN is architectural innovation for hazard-aware scene understanding. The architecture is designed to explicitly model the two-level structure of safety reasoning — first determining whether a region is dangerous, then classifying its specific type — rather than to maximise raw mIoU on a balanced benchmark. The mIoU metric is reported for comparability but should be interpreted alongside the architectural ablation study, which provides the primary evidence of contribution.

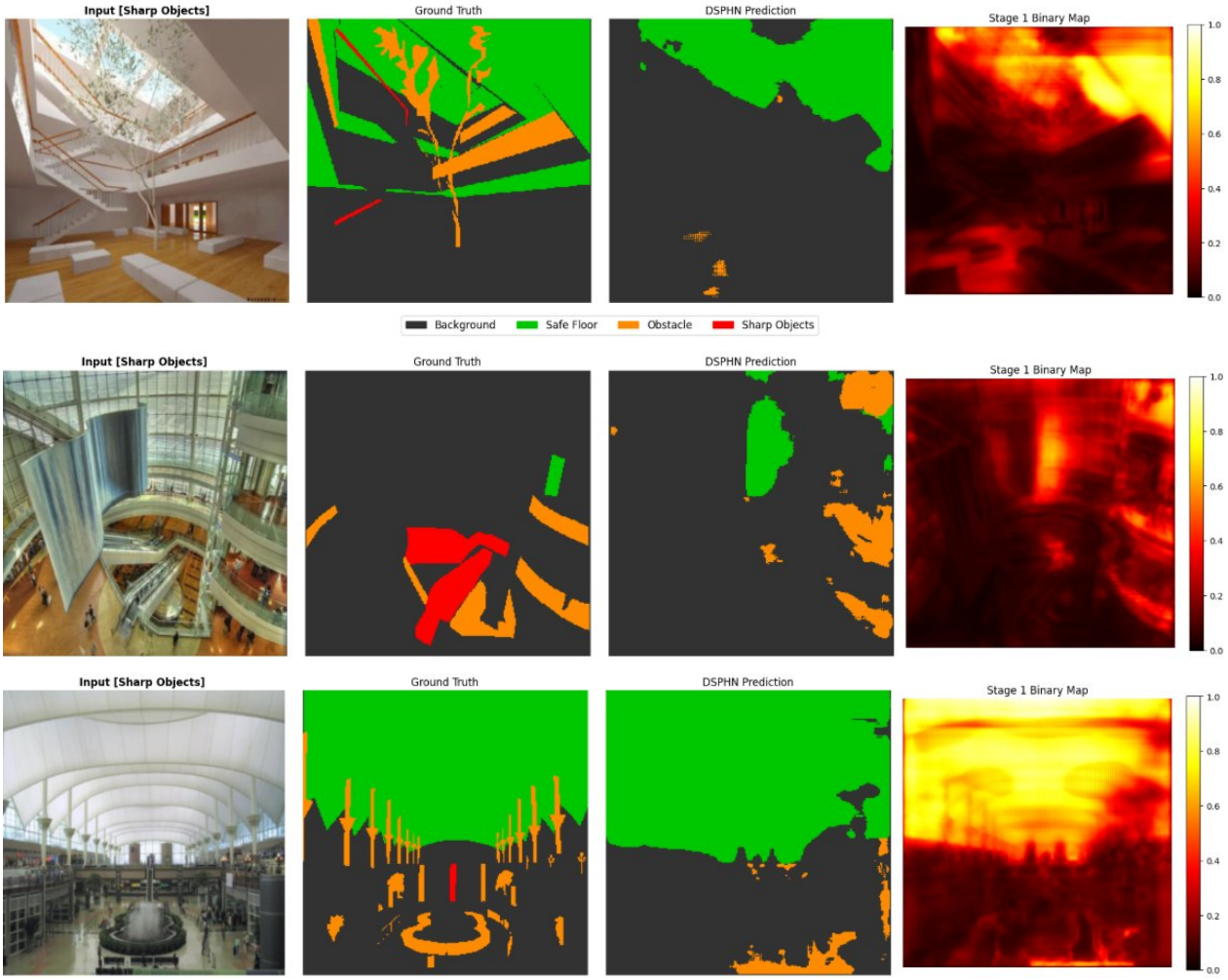


Figure 3. Qualitative results: Input, Ground Truth, DSPHN Prediction, Stage 1 Binary Hazard Map. Stage 1 maps activate (red/yellow) over hazardous floor and obstacle regions and suppress safe background (black). Colors: Background, Safe Floor, Obstacle, Sharp Objects.

Why certain classes underperform. Background dominates at 73.9% of pixels, creating strong gradient pressure toward predicting background everywhere. Safe Floor at 16.5% receives sufficient supervision for reasonable performance (IoU 0.55). Obstacle at 6.8% receives moderate supervision (IoU 0.32). Sharp Objects at 0.06% receives insufficient pixel-level supervision despite image-level overrepresentation through COCO, because the model's loss is dominated by majority-class gradients. This analysis motivates the future directions outlined in Section 6.

Why DSPHN outperforms both ablated variants. The dual-stream decoder (DAHD) provides richer feature representations by separating appearance and context learning. The two-stage mechanism (PHRN) provides hierarchical structure. In DSPHN, Stage 1 binary maps are derived from dual-stream fused features, making them more informative than single-stream Stage 1 maps, which explains why DSPHN outperforms both parents rather than falling between them.

6. Conclusion

This paper presented DSPHN, a novel semantic segmentation architecture whose primary contribution is **architectural innovation for hazard-aware scene understanding**. Three architectural components were proposed and validated:

1. **Dual-Stream Appearance-Context Decoder:** Parallel standard and dilated convolutional decoder pathways that separately model local visual hazard cues and spatial contextual relationships, addressing the complementary nature of hazard recognition signals.
2. **Severity-Guided Adaptive Fusion Gates:** Learnable gates initialised from domain injury severity scores that blend appearance and context features, providing domain-informed fusion prior to hazard classification.
3. **Two-Stage Hazard Refinement Mechanism:** A hierarchical conditional design where Stage 1 produces a binary safe/unsafe map from dual-stream fused features, and Stage 2 performs fine-

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```
=== Global Validation ===
Class                IoU     Dice    True Pixels
--------------------------------------------------
Background         0.8115   0.8959       22896675  ✅
Safe Floor         0.5380   0.6996        4645985  ✅
Obstacle           0.2888   0.4482        1943697  ✅
Sharp Objects      0.0000   0.0000           4843  ⚠️
--------------------------------------------------
Mean               0.4096   0.5109
Pixel Accuracy: 0.8349
✅ Global results saved.

=== Global Test ===
Class                IoU     Dice    True Pixels
--------------------------------------------------
Background         0.8081   0.8938       22417974  ✅
Safe Floor         0.5517   0.7111        5000776  ✅
Obstacle           0.3195   0.4843        2055612  ✅
Sharp Objects      0.0000   0.0000          16838  ⚠️
--------------------------------------------------
Mean               0.4198   0.5223
Pixel Accuracy: 0.8332
```

Figure 4. Global evaluation on validation and test sets showing per-class IoU, Dice, and true pixel counts. Sharp Objects pixel count (16,838) confirms the extreme sparsity driving zero IoU.

grained four-class classification conditioned on Stage 1 output as a direct structural input.

A controlled ablation study under identical training conditions confirms that each component contributes measurable improvement: DSPHN (mIoU 0.4198) outperforms the dual-stream variant alone (0.3894, +0.030) and the two-stage variant alone (0.3793, +0.041). This ablation provides primary evidence of the architectural contributions independent of dataset limitations.

Severe class imbalance affected performance on rare hazard classes. Sharp Objects achieved zero IoU due to pixel-level sparsity of 0.06% of test pixels — a known challenge in long-tail segmentation that represents a dataset limitation rather than an architectural failure, as confirmed by consistent ablation improvements across all evaluated classes.

Limitations. Three limitations are acknowledged: (1) zero IoU on Sharp Objects due to extreme pixel sparsity despite class weighting and COCO augmentation; (2) 15-epoch training on a T4 GPU limits convergence; (3) 256×256 resolution may miss fine-grained boundaries of small hazard instances.

Future Research Directions.

- *Larger hazard datasets:* A dedicated indoor hazard benchmark with balanced representation of all hazard classes, particularly sharp objects and wet floor surfaces, would substantially improve rare class performance and enable more rigorous evaluation.
- *Better rare-class balancing:* Patch-level oversampling of hazard regions, combined with synthetic data augmentation for rare classes, provides a more principled approach than image-level oversampling which was found to reduce overall performance.
- *Real-time deployment:* A lightweight DSPHN variant with a MobileNet or EfficientNet-Lite encoder targeting real-time inference would enable deployment on edge devices in domestic and healthcare environments.
- *Robotics and autonomous navigation:* Integration of DSPHN into robot navigation pipelines for hazard-aware path planning represents a direct application of this work, where the Stage 1 binary map provides a fast hazard alert and Stage 2 provides detailed classification for navigation decisions.

DSPHN demonstrates that explicit hazard-aware architectural design — through dual-stream feature disentanglement, severity-guided fusion, and hierarchical safety-aware inference — can improve indoor safety-oriented scene understanding and provides a foundation for future research in safety-critical semantic segmentation.

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